[INCH-POUND]
A-A-59590/4
November 27, 2000

COMMERCIAL ITEM DESCRIPTION
SPECIFICATION SHEET

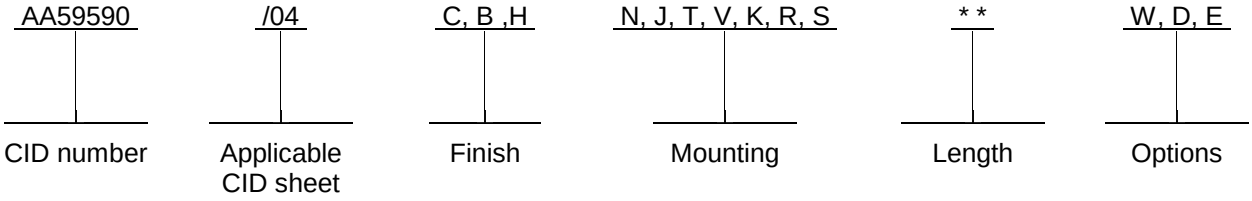
HOLDERS, ELECTRICAL CARD, WEDGE RETAINERS, 3 PIECE, SCREW ACTUATED DRIVE, .225 X .260 INCH BODY SIZE

The General Services Administration has authorized the use of this commercial item description (CID) for all federal agencies.

The complete requirements for procuring electrical card holders described herein shall consist of this document and the latest issue in effect of A-A-59590

SCOPE. This CID specification sheet covers 3 piece card holders that use screw actuated wedges to hold circuit card assemblies into their installed positions in heat sinking devices (cold plates, heat exchanger) or other applications. Card holders covered by this CID are intended to provide maximum contact between thermal paths on the circuit card assemblies and the heat sink surfaces and to provide resistance to shock and vibration. Items covered by this CID specification sheet are intended for commercial/industrial applications.

CLASSIFICATION. This CID specification sheet uses a classification system which is included in the Part Identification Number (PIN) as shown in the following example (see notes).



SALIENT CHARACTERISTICS.

Performance. Card holders shall hold the circuit card firmly in place providing high resistance to shock and vibration while providing maximum thermal transfer.

Interface and physical dimensions. The card holders supplied to this CID specification sheet shall be as specified herein and the general CID.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: Defense Supply Center, Columbus, ATTN: DSCC/VAC, Post Office Box 3990, Columbus, OH 43216-5000, or facsimile (614) 693-6939, or electronic mail 5998_Documents@dscc.dla.mil.

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Expanded and relaxed dimensions. Expanded and relaxed dimensions shall be as specified in figure 1.

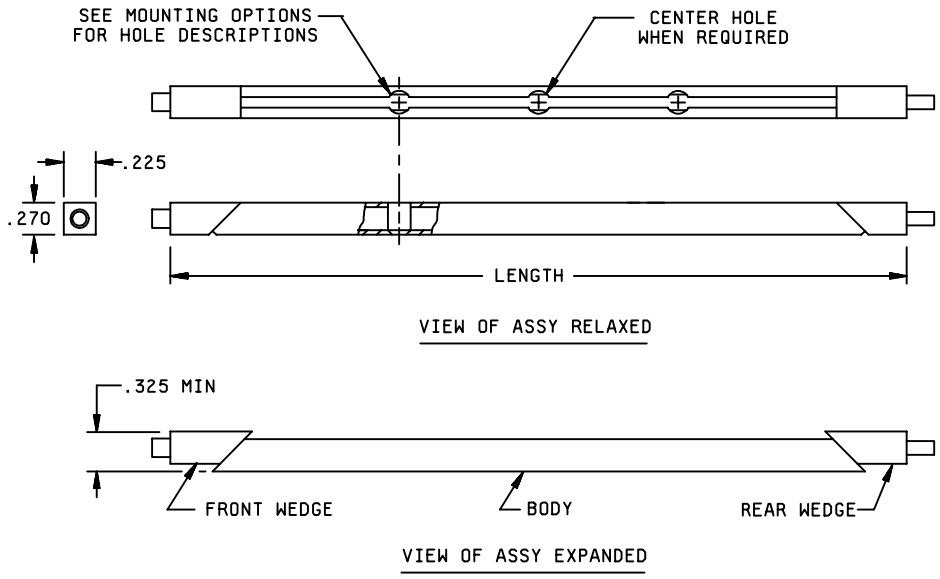

FIGURE 1 Relaxed and expanded dimensions.

Mounting options. The mounting options shall be as specified in A-A-59590.

Mounting holes (when required). Two mounting holes are required on card holders less than 6.00 inches (152.4 mm) in length. The mounting holes shall be spaced using the length of the assembly minus 1.90 inches (48.26 mm) and centered on the mounting body. Three mounting holes are required on card holders 6.00 inches or greater in length. The third mounting hole shall be centered on the mounting body.

Options. Card holders can have the following options. Card holders not requiring one of the following options shall leave the position in the PIN blank.

Lockwasher and flatwasher. A lockwasher and flatwasher located under the screw head will provide for additional resistance to loosening from shock and vibration. Card holders requiring a lockwasher and flatwasher option shall include a suffix “W” in the PIN (see classification and notes).

Screw self-locking element. The use of a screw self-locking element will provide prevailing torque for resistance to loosening from shock vibration. Card holders requiring a screw self-locking element shall include a suffix “E” in the PIN (see classification and notes).

Lockwasher, flatwasher, and screw self-locking element. Card holders requiring a lockwasher, flatwasher, and screw self-locking element option shall include a suffix “D” in the PIN (see classification and notes).

PRODUCT CONFORMANCE PROVISIONS. Product conformance shall be as specified in A-A-59590 and herein.

Performance tests. The following tests shall be performed with a 0.062 inch thick circuit card with a pair of 6 inch card holders attached. A torque of 96 ounce-inch (67.8 Newton-cm) shall be obtained to secure the assembly to a suitable test fixture.

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Vibration. Secure the card holder and circuit card assembly to a vibration system capable of producing an input of 25 Gs over a frequency range from 20 to 2000 to 20 Hz performed in 16 minutes. The assembly shall be able to withstand a vibration of 25 G-rms without dislodging or moving. This test shall be performed in both a perpendicular and parallel oscillation (in relationship to the circuit card) for a duration of 16 minutes in each orientation.

Shock. Secure the card holder and circuit card assembly to a standard drop shock tester capable of attaining at least 100 Gs. Subject the assembly to three shock pulses of 60g/6ms in each of the three principle planes (a total of 9 blows). The assembly shall be able to withstand the shock force without dislodging or moving.

Thermal transfer. With the card holder and circuit card assembly secured to a suitable cold plate test fixture, the assembly shall have a 2 degrees to 4 degrees C/W/in thermal resistance when measured over a range from 1W to 90W, dissipated by conduction only.

NOTES

PIN. The PIN should be used for government purposes to buy commercial products to this CID specification sheet. See the classification section for PIN format example.

Commercial products. As part of the market analysis and research effort, this CID specification sheet was coordinated with the following manufacturers of commercial products. At the time of CID specification sheet preparation and coordination, these manufacturers were known to have commercial products that would meet the requirements of this CID specification sheet. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict procurement to only the manufacturers shown.)

MFGRs CAGE	MFGRs Name and Address	MGFRs contact information
18915	APW / Electronic Solutions 14100 Danielson Street Poway, CA 92064-6898	Tel: (858) 679-4550 Toll Free: (800) 854-7086 Fax: (858) 679-4555 E-mail: sales@apw-elsol.com URL: www.apw-elsol.com
52094	Calmark Corporation 4915 Walnut Grove Avenue San Gabriel, CA 91776-2021	Tel: (626) 287-0451 Fax: (626) 287-7350 E-mail: sales@calmark-corp.com URL: www.calmark-corp.com
71286	Fairchild Fasteners Camloc/Ram Products Division 3000 W. Lomita Blvd. Torrance, CA 90505-5103	Tel: (310) 784-2662 Fax: (310) 784-2628 E-mail: altedesco@fairchildfasteners.com URL: www.fairchildfasteners.com

Part number (P/N) supersession data. This CID specification sheet supersedes the following manufacturer's P/Ns as shown. This information is being provided to assist in reducing proliferation in the government inventory system.

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TABLE I. Commercial P/N supersession data.

PIN designator AA59590/04	Vendor similar designator or type part number 1/ CAGE 52094	PIN designator AA59590/04	Vendor similar designator or type part number 1/ CAGE 52094	PIN designator AA59590/04	Vendor similar designator or type part number 1/ CAGE 52094
CN28	225-2.80	BN28W	WA225-2.80	HN28E	HA225-2.80L
CN30	225-3.00	BN30W	WA225-3.00	HN30E	HA225-3.00L
CN35	225-3.50	BN35W	WA225-3.50	HN35E	HA225-3.50L
CN38	225-3.80	BN38W	WA225-3.80	HN38E	HA225-3.80L
CN48	225-4.80	BN48W	WA225-4.80	HN48E	HA225-4.80L
CN53	225-5.30	BN53W	WA225-5.30	HN53E	HA225-5.30L
CN55	225-5.50	BN55W	WA225-5.50	HN55E	HA225-5.50L
CN58	225-5.80	BN58W	WA225-5.80	HN58E	HA225-5.80L
CN61	225-6.10	BN61W	WA225-6.10	HN61E	HA225-6.10L
CN67	225-6.70	BN67W	WA225-6.70	HN67E	HA225-6.70L
CJ28	225-2.80H	BJ28W	WA225-2.80H	HJ28E	HA225-2.80HL
CJ30	225-3.00H	BJ30W	WA225-3.00H	HJ30E	HA225-3.00HL
CJ35	225-3.50H	BJ35W	WA225-3.50H	HJ35E	HA225-3.50HL
CJ38	225-3.80H	BJ38W	WA225-3.80H	HJ38E	HA225-3.80HL
CJ48	225-4.80H	BJ48W	WA225-4.80H	HJ48E	HA225-4.80HL
CJ53	225-5.30H	BJ53W	WA225-5.30H	HJ53E	HA225-5.30HL
CJ55	225-5.50H	BJ55W	WA225-5.50H	HJ55E	HA225-5.50HL
CJ58	225-5.80H	BJ58W	WA225-5.80H	HJ58E	HA225-5.80HL
CK61	225-6.10EH	BK61W	WA225-6.10EH	HK61E	HA225-6.10EHL
CK67	225-6.70EH	BK67W	WA225-6.70EH	HK67E	HA225-6.70EHL
CT28	225-2.80T0	BT28W	WA225-2.80T0	HT28E	HA225-2.80T0L
CT30	225-3.00T0	BT30W	WA225-3.00T0	HT30E	HA225-3.00T0L
CT35	225-3.50T0	BT35W	WA225-3.50T0	HT35E	HA225-3.50T0L
CT38	225-3.80T0	BT38W	WA225-3.80T0	HT38E	HA225-3.80T0L
CT48	225-4.80T0	BT48W	WA225-4.80T0	HT48E	HA225-4.80T0L
CT53	225-5.30T0	BT53W	WA225-5.30T0	HT53E	HA225-5.30T0L
CT55	225-5.50T0	BT55W	WA225-5.50T0	HT55E	HA225-5.50T0L
CT58	225-5.80T0	BT58W	WA225-5.80T0	HT58E	HA225-5.80T0L
CR61	225-6.10ET0	BR61W	WA225-6.10ET0	HR61E	HA225-6.10ET0L
CR67	225-6.70ET0	BR67W	WA225-6.70ET0	HR67E	HA225-6.70ET0L

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CV28	225-2.80T2	BV28W	WA225-2.80T2	HV28E	HA225-2.80T2L
CV30	225-3.00T2	BV30W	WA225-3.00T2	HV30E	HA225-3.00T2L
CV35	225-3.50T2	BV35W	WA225-3.50T2	HV35E	HA225-3.50T2L
CV38	225-3.80T2	BV38W	WA225-3.80T2	HV38E	HA225-3.80T2L
CV48	225-4.80T2	BV48W	WA225-4.80T2	HV48E	HA225-4.80T2L
CV53	225-5.30T2	BV53W	WA225-5.30T2	HV53E	HA225-5.30T2L
CV55	225-5.50T2	BV55W	WA225-5.50T2	HV55E	HA225-5.50T2L
CV58	225-5.80T2	BV58W	WA225-5.80T2	HV58E	HA225-5.80T2L
CS61	225-6.10ET2	BS61W	WA225-6.10ET2	HS61E	HA225-6.10ET2L
CS67	225-6.70ET2	BS67W	WA225-6.70ET2	HS67E	HA225-6.70ET2L
BN28	A225-2.80	HN28W	WHA225-2.80	CN28D	W225-2.80L
BN30	A225-3.00	HN30W	WHA225-3.00	CN30D	W225-3.00L
BN35	A225-3.50	HN35W	WHA225-3.50	CN35D	W225-3.50L
BN38	A225-3.80	HN38W	WHA225-3.80	CN38D	W225-3.80L
BN48	A225-4.80	HN48W	WHA225-4.80	CN48D	W225-4.80L
BN53	A225-5.30	HN53W	WHA225-5.30	CN53D	W225-5.30L
BN55	A225-5.50	HN55W	WHA225-5.50	CN55D	W225-5.50L
BN58	A225-5.80	HN58W	WHA225-5.80	CN58D	W225-5.80L
BN61	A225-6.10	HN61W	WHA225-6.10	CN61D	W225-6.10L
BN67	A225-6.70	HN67W	WHA225-6.70	CN67D	W225-6.70L
BJ28	A225-2.80H	HJ28W	WHA225-2.80H	CJ28D	W225-2.80HL
BJ30	A225-3.00H	HJ30W	WHA225-3.00H	CJ30D	W225-3.00HL
BJ35	A225-3.50H	HJ35W	WHA225-3.50H	CJ35D	W225-3.50HL
BJ38	A225-3.80H	HJ38W	WHA225-3.80H	CJ38D	W225-3.80HL
BJ48	A225-4.80H	HJ48W	WHA225-4.80H	CJ48D	W225-4.80HL
BJ53	A225-5.30H	HJ53W	WHA225-5.30H	CJ53D	W225-5.30HL
BJ55	A225-5.50H	HJ55W	WHA225-5.50H	CJ55D	W225-5.50HL
BJ58	A225-5.80H	HJ58W	WHA225-5.80H	CJ58D	W225-5.80HL
BK61	A225-6.10EH	HK61W	WHA225-6.10EH	CK61D	W225-6.10EHL
BK67	A225-6.70EH	HK67W	WHA225-6.70EH	CK67D	W225-6.70EHL
BT28	A225-2.80T0	HT28W	WHA225-2.80T0	CT28D	W225-2.80T0L
BT30	A225-3.00T0	HT30W	WHA225-3.00T0	CT30D	W225-3.00T0L
BT35	A225-3.50T0	HT35W	WHA225-3.50T0	CT35D	W225-3.50T0L
BT38	A225-3.80T0	HT38W	WHA225-3.80T0	CT38D	W225-3.80T0L
BT48	A225-4.80T0	HT48W	WHA225-4.80T0	CT48D	W225-4.80T0L
BT53	A225-5.30T0	HT53W	WHA225-5.30T0	CT53D	W225-5.30T0L
BT55	A225-5.50T0	HT55W	WHA225-5.50T0	CT55D	W225-5.50T0L
BT58	A225-5.80T0	HT58W	WHA225-5.80T0	CT58D	W225-5.80T0L
BR61	A225-6.10ET0	HR61W	WHA225-6.10ET0	CR61D	W225-6.10ET0L
BR67	A225-6.70ET0	HR67W	WHA225-6.70ET0	CR67D	W225-6.70ET0L

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BV28	A225-2.80T2	HV28W	WHA225-2.80T2	CV28D	W225-2.80T2L
BV30	A225-3.00T2	HV30W	WHA225-3.00T2	CV30D	W225-3.00T2L
BV35	A225-3.50T2	HV35W	WHA225-3.50T2	CV35D	W225-3.50T2L
BV38	A225-3.80T2	HV38W	WHA225-3.80T2	CV38D	W225-3.80T2L
BV48	A225-4.80T2	HV48W	WHA225-4.80T2	CV48D	W225-4.80T2L
BV53	A225-5.30T2	HV53W	WHA225-5.30T2	CV53D	W225-5.30T2L
BV55	A225-5.50T2	HV55W	WHA225-5.50T2	CV55D	W225-5.50T2L
BV58	A225-5.80T2	HV58W	WHA225-5.80T2	CV58D	W225-5.80T2L
BS61	A225-6.10ET2	HS61W	WHA225-6.10ET2	CS61D	W225-6.10ET2L
BS67	A225-6.70ET2	HS67W	WHA225-6.70ET2	CS67D	W225-6.70ET2L
HN28	HA225-2.80	CN28E	225-2.80L	BN28D	WA225-2.80L
HN30	HA225-3.00	CN30E	225-3.00L	BN30D	WA225-3.00L
HN35	HA225-3.50	CN35E	225-3.50L	BN35D	WA225-3.50L
HN38	HA225-3.80	CN38E	225-3.80L	BN38D	WA225-3.80L
HN48	HA225-4.80	CN48E	225-4.80L	BN48D	WA225-4.80L
HN53	HA225-5.30	CN53E	225-5.30L	BN53D	WA225-5.30L
HN55	HA225-5.50	CN55E	225-5.50L	BN55D	WA225-5.50L
HN58	HA225-5.80	CN58E	225-5.80L	BN58D	WA225-5.80L
HN61	HA225-6.10	CN61E	225-6.10L	BN61D	WA225-6.10L
HN67	HA225-6.70	CN67E	225-6.70L	BN67D	WA225-6.70L
HJ28	HA225-2.80H	CJ28E	225-2.80HL	BJ28D	WA225-2.80HL
HJ30	HA225-3.00H	CJ30E	225-3.00HL	BJ30D	WA225-3.00HL
HJ35	HA225-3.50H	CJ35E	225-3.50HL	BJ35D	WA225-3.50HL
HJ38	HA225-3.80H	CJ38E	225-3.80HL	BJ38D	WA225-3.80HL
HJ48	HA225-4.80H	CJ48E	225-4.80HL	BJ48D	WA225-4.80HL
HJ53	HA225-5.30H	CJ53E	225-5.30HL	BJ53D	WA225-5.30HL
HJ55	HA225-5.50H	CJ55E	225-5.50HL	BJ55D	WA225-5.50HL
HJ58	HA225-5.80H	CJ58E	225-5.80HL	BJ58D	WA225-5.80HL
HK61	HA225-6.10EH	CK61E	225-6.10EHL	BK61D	WA225-6.10EHL
HK67	HA225-6.70EH	CK67E	225-6.70EHL	BK67D	WA225-6.70EHL
HT28	HA225-2.80T0	CT28E	225-2.80T0L	BT28D	WA225-2.80T0L
HT30	HA225-3.00T0	CT30E	225-3.00T0L	BT30D	WA225-3.00T0L
HT35	HA225-3.50T0	CT35E	225-3.50T0L	BT35D	WA225-3.50T0L
HT38	HA225-3.80T0	CT38E	225-3.80T0L	BT38D	WA225-3.80T0L
HT48	HA225-4.80T0	CT48E	225-4.80T0L	BT48D	WA225-4.80T0L
HT53	HA225-5.30T0	CT53E	225-5.30T0L	BT53D	WA225-5.30T0L
HT55	HA225-5.50T0	CT55E	225-5.50T0L	BT55D	WA225-5.50T0L
HT58	HA225-5.80T0	CT58E	225-5.80T0L	BT58D	WA225-5.80T0L
HR61	HA225-6.10ET0	CR61E	225-6.10ET0L	BR61D	WA225-6.10ET0L
HR67	HA225-6.70ET0	CR67E	225-6.70ET0L	BR67D	WA225-6.70ET0L

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HV28	HA225-2.80T2	CV28E	225-2.80T2L	BV28D	WA225-2.80T2L
HV30	HA225-3.00T2	CV30E	225-3.00T2L	BV30D	WA225-3.00T2L
HV35	HA225-3.50T2	CV35E	225-3.50T2L	BV35D	WA225-3.50T2L
HV38	HA225-3.80T2	CV38E	225-3.80T2L	BV38D	WA225-3.80T2L
HV48	HA225-4.80T2	CV48E	225-4.80T2L	BV48D	WA225-4.80T2L
HV53	HA225-5.30T2	CV53E	225-5.30T2L	BV53D	WA225-5.30T2L
HV55	HA225-5.50T2	CV55E	225-5.50T2L	BV55D	WA225-5.50T2L
HV58	HA225-5.80T2	CV58E	225-5.80T2L	BV58D	WA225-5.80T2L
HS61	HA225-6.10ET2	CS61E	225-6.10ET2L	BS61D	WA225-6.10ET2L
HS67	HA225-6.70ET2	CS67E	225-6.70ET2L	BS67D	WA225-6.70ET2L
CN28W	W225-2.80	BN28E	A225-2.80L	HN28D	WHA225-2.80L
CN30W	W225-3.00	BN30E	A225-3.00L	HN30D	WHA225-3.00L
CN35W	W225-3.50	BN35E	A225-3.50L	HN35D	WHA225-3.50L
CN38W	W225-3.80	BN38E	A225-3.80L	HN38D	WHA225-3.80L
CN48W	W225-4.80	BN48E	A225-4.80L	HN48D	WHA225-4.80L
CN53W	W225-5.30	BN53E	A225-5.30L	HN53D	WHA225-5.30L
CN55W	W225-5.50	BN55E	A225-5.50L	HN55D	WHA225-5.50L
CN58W	W225-5.80	BN58E	A225-5.80L	HN58D	WHA225-5.80L
CN61W	W225-6.10	BN61E	A225-6.10L	HN61D	WHA225-6.10L
CN67W	W225-6.70	BN67E	A225-6.70L	HN67D	WHA225-6.70L
CJ28W	W225-2.80H	BJ28E	A225-2.80HL	HJ28D	WHA225-2.80HL
CJ30W	W225-3.00H	BJ30E	A225-3.00HL	HJ30D	WHA225-3.00HL
CJ35W	W225-3.50H	BJ35E	A225-3.50HL	HJ35D	WHA225-3.50HL
CJ38W	W225-3.80H	BJ38E	A225-3.80HL	HJ38D	WHA225-3.80HL
CJ48W	W225-4.80H	BJ48E	A225-4.80HL	HJ48D	WHA225-4.80HL
CJ53W	W225-5.30H	BJ53E	A225-5.30HL	HJ53D	WHA225-5.30HL
CJ55W	W225-5.50H	BJ55E	A225-5.50HL	HJ55D	WHA225-5.50HL
CJ58W	W225-5.80H	BJ58E	A225-5.80HL	HJ58D	WHA225-5.80HL
CK61W	W225-6.10EH	BK61E	A225-6.10EHL	HK61D	WHA225-6.10EHL
CK67W	W225-6.70EH	BK67E	A225-6.70EHL	HK67D	WHA225-6.70EHL
CT28W	W225-2.80T0	BT28E	A225-2.80T0L	HT28D	WHA225-2.80T0L
CT30W	W225-3.00T0	BT30E	A225-3.00T0L	HT30D	WHA225-3.00T0L
CT35W	W225-3.50T0	BT35E	A225-3.50T0L	HT35D	WHA225-3.50T0L
CT38W	W225-3.80T0	BT38E	A225-3.80T0L	HT38D	WHA225-3.80T0L
CT48W	W225-4.80T0	BT48E	A225-4.80T0L	HT48D	WHA225-4.80T0L
CT53W	W225-5.30T0	BT53E	A225-5.30T0L	HT53D	WHA225-5.30T0L
CT55W	W225-5.50T0	BT55E	A225-5.50T0L	HT55D	WHA225-5.50T0L
CT58W	W225-5.80T0	BT58E	A225-5.80T0L	HT58D	WHA225-5.80T0L
CR61W	W225-6.10ET0	BR61E	A225-6.10ET0L	HR61D	WHA225-6.10ET0L
CR67W	W225-6.70ET0	BR67E	A225-6.70ET0L	HR67D	WHA225-6.70ET0L

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CV28W	W225-2.80T2	BV28E	A225-2.80T2L	HV28D	WHA225-2.80T2L
CV30W	W225-3.00T2	BV30E	A225-3.00T2L	HV30D	WHA225-3.00T2L
CV35W	W225-3.50T2	BV35E	A225-3.50T2L	HV35D	WHA225-3.50T2L
CV38W	W225-3.80T2	BV38E	A225-3.80T2L	HV38D	WHA225-3.80T2L
CV48W	W225-4.80T2	BV48E	A225-4.80T2L	HV48D	WHA225-4.80T2L
CV53W	W225-5.30T2	BV53E	A225-5.30T2L	HV53D	WHA225-5.30T2L
CV55W	W225-5.50T2	BV55E	A225-5.50T2L	HV55D	WHA225-5.50T2L
CV58W	W225-5.80T2	BV58E	A225-5.80T2L	HV58D	WHA225-5.80T2L
CS61W	W225-6.10ET2	BS61E	A225-6.10ET2L	HS61D	WHA225-6.10ET2L
CS67W	W225-6.70ET2	BS67E	A225-6.70ET2L	HS67D	WHA225-6.70ET2L

1/ The manufacturer's P/N shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph.

MILITARY INTERESTS:

Custodians:
Air Force - 11
Army - CR
DLA - CC
Navy - EC

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - 7FXE

Preparing Activity

DLA - CC

Project 5998-0097-04

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